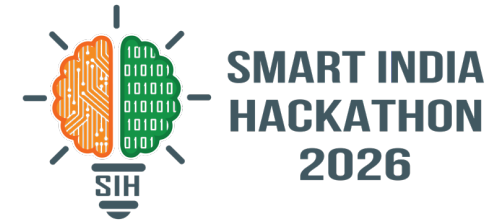


SMART INDIA HACKATHON 2026



Problem Statement ID – SIH25002

Problem Statement Title – Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID

Theme – Travel & Tourism

PS Category – Software

Organisation – Ministry of Development of North Eastern Region (MDoNER)

Team Name – STS Safety



Working prototype · tourist and command surfaces

LIVE PRODUCT · ₹0 INFRA

1. The problem — North-East tourist safety

Tourism is national-scale. Coverage is not.

- 1.21 crore domestic + 2.21 lakh foreign visits to the 8 NE states in 2023 (PIB / State Tourism Depts).
- NE share of India's foreign tourist arrivals is only 2.05% (1.95 lakh; MoT Compendium 2024) — a priority region that is still under-visited.
- 7 of 8 NE states have no dedicated Tourist Police. MoT's 15-state list includes Sikkim only (Lok Sabha 27.07.2026).
- NCRB does not maintain crime data on domestic tourists. 156 cases against foreign tourists were registered in 2024 (NCRB / LS AU1299) — missing-person, restricted-zone and signal-loss events are invisible.
- Check-post, hotel, forest office and police do not share a database. A Tawang checkpoint cannot verify a Guwahati-issued ID if the issuing server is down.

PS ask: digital ID + geo-fence + AI + SOS + E-FIR — with DPDP-safe privacy.

2. Three-layer system (the prototype)

- 01 Geo-fence in Postgres**
AFTER INSERT trigger + PostGIS ST_Covers / ST_DWithin. The phone cannot skip an incident. Debounce via partial unique index so a border-stander does not generate 400 alerts.
- 02 Soulbound digital ID**
ERC-5192 locked TDID on Polygon Amoy. Chain stores only keccak256(KYC || salt) + itinerary hash. KYC stays pgcrypto-encrypted in Postgres. Selective disclosure at /verify.
- 03 AI beside the path — never on it**
Tier-1 SQL rules always fire. IsolationForest only ranks. Groq/Gemini write the brief & E-FIR draft. If HF Space, Groq and Amoy are all down, SOS still lands.

3. What no typical build does

- ▶ **Safety path is the database.**
Blockchain and AI sit beside it. Unplug the WAN: Realtime + Anvil + ONNX + PMTiles still demo.
- ▶ **On-device warning first.**
Turf.js evaluates cached zones before the packet arrives. IndexedDB + Background Sync + sms: fallback for Arunachal towers.
- ▶ **Lawful identity, not an NFT.**
Zero PII on-chain. Hotel learns match / no-match, never the passport number. Expired trip auto-invalidates on-chain.
- ▶ **Evidentiary E-FIR, not a chatbot FIR.**
Officer must approve. PDF hash is anchored. LLM never files to CCTNS.
- ▶ **Honest ML.**
Trained on synthetic NE trajectories (DPDP). Hold-out: 0 FP on 1,500 normal-trek windows. Rules catch what the forest misses.

180–450 ms

SOS → first channel
measured, local stack

0 FP / 1,500

normal-trek hold-out
IsolationForest; rules still decide

₹0 infra

no credit card in the stack
Vercel + Supabase + HF + Amoy

5 locales

en · hi · as · bn · ne
PS multilingual requirement

< 2 s

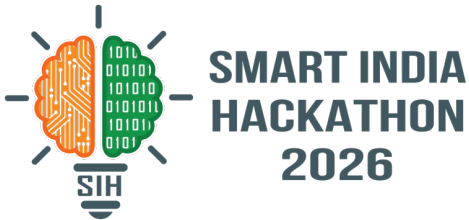
Telegram target (online)
panic path skips debounce

Sources (problem stats): PIB 'Tourism in North-Eastern Region' Dec 2024; MoT India Tourism Data Compendium Key Highlights 2024; Lok Sabha USQ 1299, 27.07.2026 (NCRB). Product stats: repo README + services/ai/artifacts/metrics.json.



TECHNICAL APPROACH

Working prototype



Technologies — every row is free-tier, no card

Layer	Choice	Why this, not the obvious
App	Next.js 16 PWA + MapLibre	URL, not an APK. Web Push + Background Sync included.
Hot path	PostgREST insert + RLS	Skips serverless. 10 writes/s is noise for Postgres.
Geo engine	PostGIS 3 · plpgsql trigger	ST_Covers in the database. Client cannot bypass.
Maps	OpenFreeMap + Turf.js	No Mapbox/Google key. On-device pre-check.
Identity	Solidity 0.8.24 · ERC-5192	Soulbound TDID. keccak256 commitment only.
Chain	Polygon Amoy / Anvil	Free POL. Offline = Foundry Anvil.
ML	IsolationForest → ONNX	HF Space + onnxruntime-node fallback.
LLM	Groq → Gemini → template	Brief + E-FIR draft. Never gates the alert.
Notify	Realtime + VAPID + Telegram	Twilio/SMS is a stub for a ministry short-code.
Privacy	pgcrypto + RLS + 24h purge	DPDP: consent, purpose-limit, right to erasure.

Methodology — SOS / geo-fence hot path

1

Device

watchPosition
Turf.js local zone
IndexedDB if offline

2

Ingest

RLS insert into
location_pings
~80–150 ms

3

PostGIS

evaluate_position()
zone · corridor ·
speed · silence

4

Incident

INSERT incidents
Realtime <100 ms
control-room siren

5

Side path

ONNX score
LLM brief
chain anchor

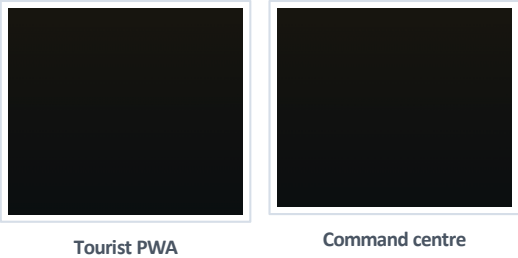
6

Dispatch

nearest unit
Telegram + Push
E-FIR draft

Panic path — button to siren

Hold 1.5 s → direct RLS incident insert → Realtime takeover
→ Telegram + Push + email in parallel.
Offline: Background Sync + pre-composed SMS with
coordinates.
AI and blockchain stay beside the safety path.



Surfaces: /home /sos /onboard /dashboard /verify · Simulator stands in for the IoT band on the same ingest path.

Feasibility — already built and measured

Hold-out IsolationForest	P = 1.00 R = 0.56 F1 = 0.72 n_test = 1,650 · threshold 0.72 · 18 features
Confusion (hold-out)	TN 1,500 · FP 0 · TP 84 · FN 66 0 false positives on 1,500 normal-trek windows
Scenario recall	zone-breach 1.00 · stationary-anomaly 1.00 route-deviation 0.00 — caught by PostGIS rules, not the forest
SOS latency	180–450 ms first channel (local) Telegram target < 2,000 ms when the bot is live
Cost / scale	₹0 at demo · 24 h ping downsample 50 tourists × 1 ping / 5 s ≈ 104 MB/day before downsample

Viability — government adoption, not a startup deck

Who pays: MDoNER / state tourism + police. We are a processor; they are the data fiduciary.

Why they would: 7 NE states have no Tourist Police. This is a force-multiplier for 112 / control rooms, not a consumer app.

Month-one production (if funded): (1) ministry SMS short-code, (2) real ILP/KYC adapter, (3) non-pausing Postgres, (4) Amony → Polygon mainnet with a hardware-backed issuer, (5) CERT-In threat model. The geofence engine does not change.

We do not claim a ₹-crore TAM or a 20% crime drop. Those numbers are not in any official series we can cite.

Challenges, risks, and the strategy already in the repo

01

**Low connectivity
(Arunachal / hills)**

IndexedDB queue + Background Sync. Turf.js warns on-device. SOS falls back to sms: with coordinates. Offline demo: four env switches (DB / chain / AI / tiles).

02

**GPS spoof / rooted
phone**

Cannot stop a rooted device. Refuse implausible speed (>150 km/h). Session JWT. SOS is higher-trust because it is a user gesture. Checkpoint QR still binds the person.

03

**Synthetic ML data
(DPDP)**

Stated in the pitch. IsolationForest is unsupervised on generated NE tracks. The decision to alert is rules + PostGIS. Rules-only is one env flip.

04

LLM + legal E-FIR

Correct: an LLM cannot file. We draft a narrative PDF. approved_by is a human. Nothing is submitted to CCTNS from this system.

05

**Free-tier pause /
rate-limit on stage**

GitHub Action hits /api/health every 6 h. Local supabase start + Anvil + ONNX + PMTiles is the stage fallback. Rehearsed unplugged.

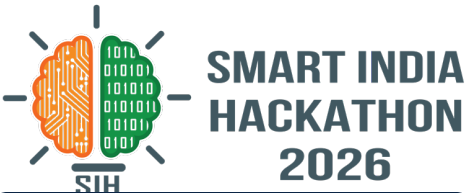
06

**Privacy / DPDP /
“is this surveillance?”**

Opt-in, trip-scoped, purpose = safety. KYC encrypted, never on-chain, never logged. RLS on every table. 24 h ping retention. Emergency contacts only at critical.



IMPACT AND BENEFITS



Social — who is safer

- Tourists (domestic + foreign) in Kaziranga, Tawang, Sohra, Loktak, border approaches — restricted-zone warning before they are lost.
- Women / solo travellers: hold-to-SOS that actually reaches a human, with last-known point and digital ID.
- Families: opt-in tracking, emergency contacts only at critical — not a live broadcast.
- Check-post / hotel / forest staff: one QR, green/amber/red, no central API dependency.
- 5 UI languages (en, hi, as, bn, ne) matching the PS multilingual clause.

Economic — what it unlocks

- Tourism is 5.22% of GDP and 84.63 million jobs. NE is a MoT + MDoNER priority that still takes only 2.05% of FTAs.
- Perception of safety is a binding constraint on high-value / foreign itineraries into the hills.
- Control-room MTTA/MTTR becomes measurable. Today most NE states have no Tourist Police unit to even record that clock.
- ₹0 to run the prototype. First paid line in production is SMS + a non-pausing database — not a rewrite.
- E-FIR drafts cut officer paperwork; the officer still signs.

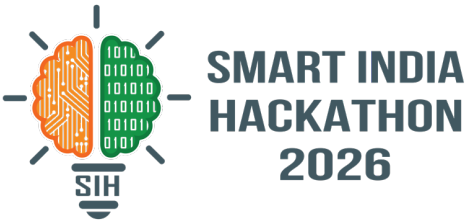
Governance / environment

- MDoNER PS delivered as specified: ID + geo-fence + AI + dashboard + E-FIR.
- Aligns with Tourist Police Scheme (BPR&D / MHA / MoT national conference) — a software layer those 7 NE states can actually use.
- DPDP 2023: data minimisation (no PII on-chain), purpose limitation, retention cron, right to erasure.
- Kaziranga / forest restricted polygons reduce illegal entry into core habitat — a conservation side-effect of the geo-fence, not a claimed ‘carbon’ product.
- Inner Line / LAC: we seed public caution polygons (Bum La approach), not classified overlays.

We do not invent a ‘lives saved’ or ‘₹ X crore unlocked’ figure. Impact is: faster, auditable incident response for a region that currently has neither a tourist-police layer nor a domestic-tourist crime series.



RESEARCH AND REFERENCES



Comparison with existing / typical SIH tourist-safety builds

Capability	STS Safety (this prototype)	Typical conventional build
Geo-fence	PostGIS trigger; client cannot skip	App if-else or Mongo \$geoWithin
Digital ID	ERC-5192 soulbound + keccak256 commitment	QR in a database, or NFT with PII
AI role	Ranks only. Rules decide. LLM never gates SOS	LLM on GPS / 'predictive hotspot'
Offline	Turf.js + IndexedDB + sms: + Anvil + ONNX	Needs internet / paid Map SDK
Privacy	RLS, pgcrypto, 24 h ping purge, no PII on-chain	Central Aadhaar log
E-FIR	Officer-approved PDF; hash anchored	Chatbot 'files' a legal document
Cost	₹0, no card, four local fallbacks	Twilio + Mapbox + Firebase Blaze
ML honesty	Synthetic tracks; metrics published	Unstated accuracy on unseen GPS

Future scope — only what the architecture already allows

- Ministry SMS short-code on the existing INotificationChannel adapter (sms.stub.ts).
- Real Inner Line Permit / KYC adapter; custodial HD wallet → ERC-4337 paymaster.
- Polygon mainnet + hardware-backed issuer. Contracts are vanilla OZ v5 — config, not a rewrite.
- Physical IoT band posting to the same location_pings ingest (the simulator is that interface today).
- CERT-In intake + threat model. Geofence engine stays in Postgres.

References — clickable in the PDF

- MoT / PIB — Tourism in NER, DTV+FTV 2023
- MoT Compendium Key Highlights 2024 (NE FTA 2.05%)
- PIB — FTA 9.95M (2024); GDP 5.22%; jobs 84.63M
- Lok Sabha USQ 1299 (27.07.2026) — Tourist Police + NCRB 156
- DPDP Act 2023 · ERC-5192 · W3C Verifiable Credentials
- Liu et al., Isolation Forest, ICDM 2008 · PostGIS ST_Covers / GiST
- SIH PS SIH25002 — MDoNER, Travel & Tourism, Software

Prototype: tourist PWA + command centre + checkpoint verify + PostGIS engine + ONNX model + Anvil/Amoy contracts. Demo script: docs/DEMO-SCRIPT.md · Hostile Q&A: docs/JUDGE-QA.md